

Sydney Technical High School



Mathematics

HSC Assessment Task 3 - 2 unit

June 2014

General Instructions

- Working Time – 75 minutes.
- Approved calculators may be used.
- Write using blue or black pen.
- A table of Standard Integrals is provided at the back of this paper.
- In Question 6 - 9, show relevant mathematical reasoning and/or calculations.
- Begin each question on a new side of the answer booklet.
- Marks shown are a guide and may need to be adjusted.
- Full marks may not be awarded for careless work or illegible writing.

NAME _____

TEACHER _____

Total Marks – 57

MULTIPLE CHOICE Pages 2 – 3

5 marks

FREE RESPONSE Pages 3 – 5

52 marks

MULTIPLE CHOICE Q1-5

Question 1

What is the derivative of $\sin^2 x$?

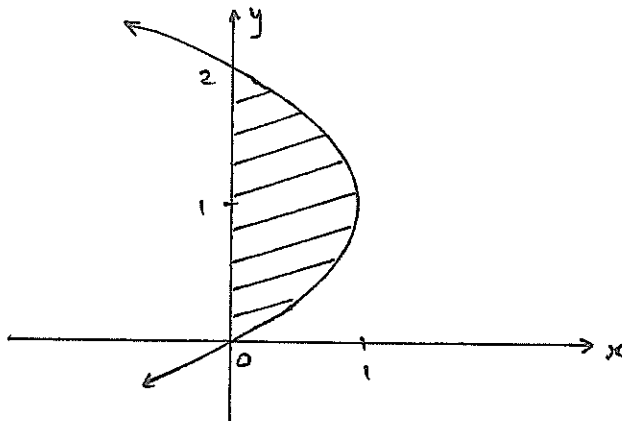
- A. $\cos^2 x$ B. $2 \sin x$ C. $2 \cos x \sin x$ D. $\frac{\sin^3 x}{3}$

Question 2

Which of the following curves has period π units and amplitude 2 units?

- A. $y = \pi \tan \frac{x}{2}$ B. $y = \pi \sin 2x$ C. $y = 2 \cos 2x$ D. $y = 2 \sin \frac{x}{2}$

Question 3

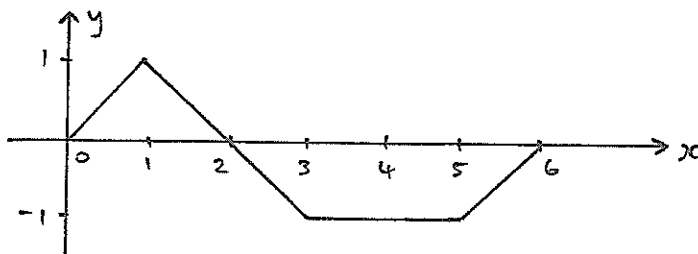


The shaded area can be found using:

- A. Area = $\int_0^1 f(y) dy$ B. Area = $\int_0^1 f(x) dx$
C. Area = $\int_0^2 f(y) dy$ D. Area = $\int_0^2 f(y) dx$

Question 4

Use the graph of $y = f(x)$ below to evaluate $\int_0^6 f(x) dx$



- A. 2 B. -2 C. -3 D. 4

Question 5

$$\frac{d}{dx}(\cot x) = ?$$

- A. $-\operatorname{cosec}^2 x$ B. $-\cos^2 x$ C. $\sec x \tan x$ D. $-\cot x \operatorname{cosec} x$
-

FREE RESPONSE Q6-9

Question 6 (13 marks)

- a) Express 10° in radians. 1
- b) Write the exact value of $\cos \frac{\pi}{6}$ 1
- c) A circle, with radius 10 cm, has a sector subtending $\frac{\pi}{5}$ radians at the centre. Find the area of the sector in exact form. 1
- d) Solve for $0 \leq x \leq 2\pi$: (answers in exact form)
- i) $\tan x = \sqrt{3}$ 2
- ii) $2\sin^2 x + \sin x = 0$ 3
- e) Differentiate:
- i) $y = \sin 3x$ 1
- ii) $y = x \cos x$ 2
- iii) $y = \sqrt{\tan x}$ 2

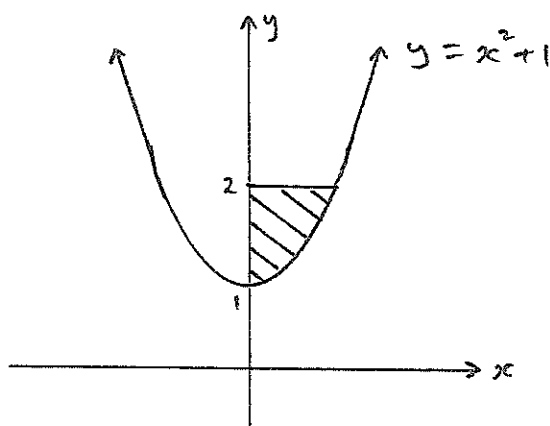
Question 7 (13 marks) Start a new page.

- a) For the curve $y = \cos 3x$:
- i) What is its period? 1
- ii) Sketch the curve over the domain $0 \leq x \leq \pi$. Use a ruler and show intercepts on the axes. 2
- b) A curve has gradient function $\frac{dy}{dx} = 3x^2 + x - 1$. If the curve passes through $(-1, 1)$, find the equation of the curve. 2
- c) Find the gradient of the curve $y = 5x + 2 \sin x$ when $x = 0$. 2
- d) If $\int_0^a (2x + 2)^3 dx = 30$, find the value of a . 2

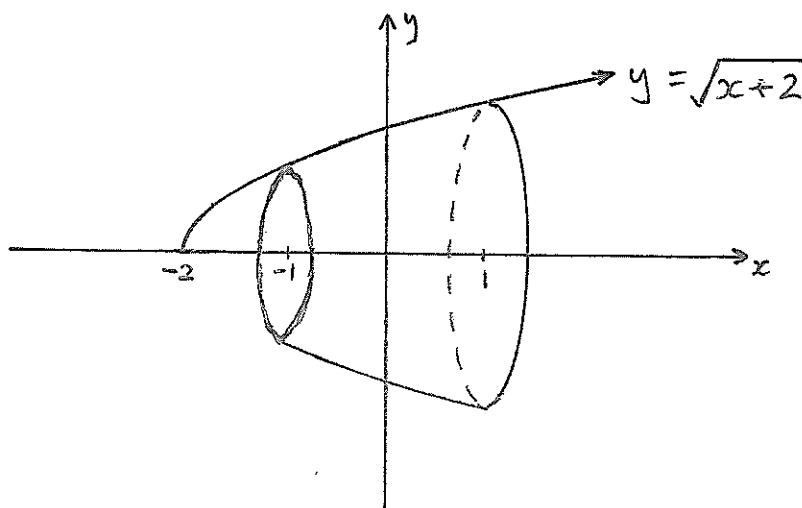
- e) Consider the curve given by $y = \sin x - \cos x$. Find the coordinates of all stationary points in the domain $0 \leq x \leq 2\pi$ and determine their nature. Do not sketch the curve. 4

Question 8 (13 marks) Start a new page.

- a) Differentiate $y = \frac{\sin 2x}{2x}$. Simplify your answer. 2
- b) i) Use two (2) applications of the Trapezoidal rule to approximate the value of $\int_1^3 \frac{1}{x} dx$ 2
- ii) Will your answer in part i) be an over OR under-estimation of the true integral value? 1
- Accurately explain why.
- c) i) Given $y = x^2 + 1$, express x in terms of y . 1
- ii) Using your answer above, find the shaded area below. 2



- iii) Derive the same shaded area above using the x axis. 2
- d) A hollow bowl is made by rotating part of the curve $y = \sqrt{x+2}$ between $x = -1$ and $x = 1$ around the x axis, as shown below. Find the exact volume occupied by the bowl. 3



Question 9 (13 marks) Start a new page.

a) Solve $\cos x = 0.7256$ for $0 \leq x \leq 2\pi$. Give your answer in radians correct to two decimal places. 2

b) i) Differentiate $\sin^4 x$ 1

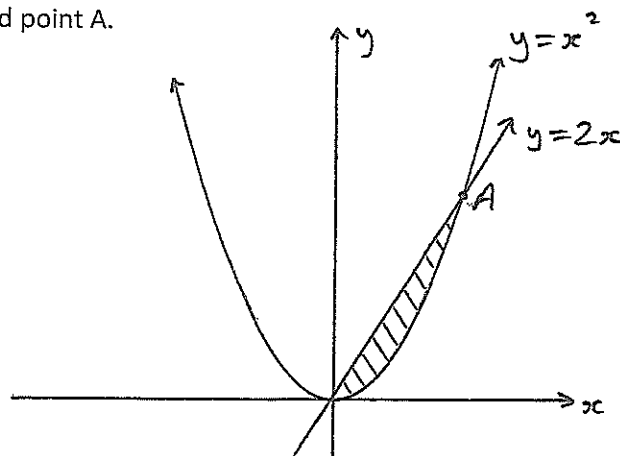
ii) Hence, find $\int \cos x \sin^3 x \, dx$ 1

c) Use Simpson's Rule and the five (5) function values in the table below to 2

approximate $\int_3^7 f(x) \, dx$

x	3	4	5	6	7
$f(x)$	2	5	1	3	4

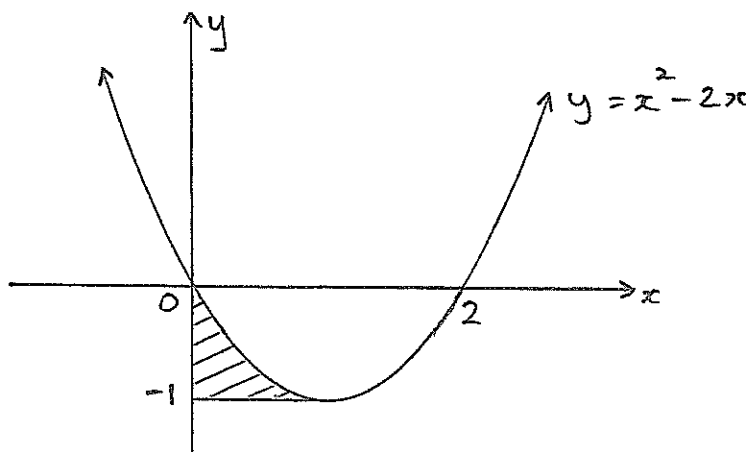
d) The area between the parabola $y = x^2$ and the line $y = 2x$ is shown. The graphs intersect at the origin and point A.



i) Find the coordinates of A. 1

ii) Find the exact value of the shaded area. 3

e) Find the shaded area below: 3



SOLUTIONS (24 yr 12)

① C ② C ③ C ④ B ⑤ A

⑥ a) $10 \times \frac{\pi}{180} = \frac{\pi}{18}$

b) $\frac{\sqrt{3}}{2}$

c) $A = \frac{1}{2} r^2 \theta$
 $= \frac{1}{2} \times 100 \times \frac{\pi}{5}$
 $= 10\pi \text{ u}^2$

d) i) $x = \frac{\pi}{3}$ (1st, 3rd quadrants)
 $= \frac{\pi}{3}, \frac{4\pi}{3}$

ii) $\sin x (2\sin x + 1) = 0$

$\sin x = 0$ or $-\frac{1}{2}$

$\therefore x = 0, \pi, 2\pi, \frac{7\pi}{6}, \frac{11\pi}{6}$

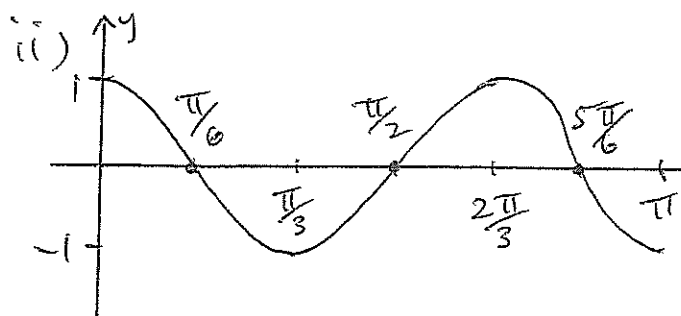
e) i) $y' = 3 \cos 3x$

ii) $y' = 1 \times \cos x + (-\sin x)x$
 $= \cos x - x \sin x$

iii) $y' = \frac{1}{2} (\tan x)^{-\frac{1}{2}} \times \sec^2 x$

or $\frac{\sec^2 x}{2\sqrt{\tan x}}$

⑦ a) i) period is $\frac{2\pi}{3}$ units



b) $y = x^3 + \frac{x^2}{2} - x + c$

$(-1, 1) \rightarrow 1 = -1 + \frac{1}{2} - 1 + c$
 $(c = \frac{1}{2})$

$\therefore y = x^3 + \frac{x^2}{2} - x + \frac{1}{2}$

c) $y' = 5 + 2 \cos x$

when $x = 0$, gradient $= 5 + 2$
 $= 7$

d) $\left[\frac{(2x+2)^4}{8} \right]_0^a = 30$

$\frac{(2a+2)^4}{8} - \frac{2^4}{8} = 30$

$\frac{(2a+2)^4}{8} = 32$

$(2a+2)^4 = 256$

$2a+2 = 4$

$\therefore a = 1$

⑦ e) S.P.'s when $y' = \cos x + \sin x = 0$

$$\therefore \sin x = -\cos x$$

$$\tan x = -1$$

$$\therefore x = \frac{\pi}{4} \text{ (2nd, 4th quads)}$$

$$= \frac{3\pi}{4} \text{ or } \frac{7\pi}{4}$$

Testing :

x	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
y'	+	0	-

 $\Rightarrow$ max. turning pt. at $(\frac{3\pi}{4}, \sqrt{2})$

x	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
y'	-	0	+

 $\Rightarrow$ min. turning pt. at $(\frac{7\pi}{4}, -\sqrt{2})$

⑧ a) $y = \frac{2 \cos 2x \times 2x - 2 \sin 2x}{4x^2}$

$$= \frac{2x \cos 2x - \sin 2x}{2x^2}$$

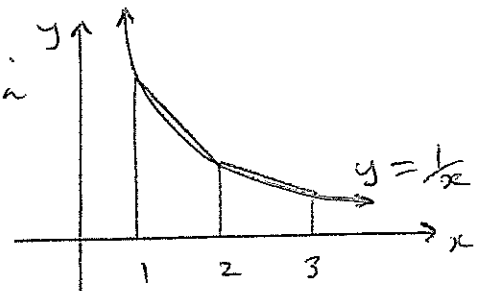
b) i) $\int_1^3 \frac{1}{x} dx \doteq \frac{1}{2} [f(1) + f(2)] + \frac{1}{2} [f(2) + f(3)]$

$$= \frac{1}{2} (1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{3})$$

$$= \frac{1}{2} (2\frac{1}{3})$$

$$= 1\frac{1}{6}$$

ii) answer is an over estimation
as the trapeziums go above
the curve



c) i) $x^2 = y - 1$

$$\therefore x = \pm \sqrt{y-1}$$

ii) Area = $\int_1^2 (y-1)^{1/2} dy$

$$= \left[\frac{2}{3} (y-1)^{3/2} \right]_1^2 = \frac{2}{3} (1-0) = \frac{2}{3} u^2$$

$$\begin{aligned}
 (8) \text{ c) iii) Area} &= 2 \times 1 (\text{rectangle}) - \int_0^1 (x^2 + 1) dx \\
 &= 2 - \left[\frac{x^3}{3} + x \right]_0^1 \\
 &= 2 - \left[\left(\frac{1}{3} + 1 \right) - (0 + 0) \right] \\
 &= 2 - 1\frac{1}{3} \\
 &= \frac{2}{3} \text{ as reqd.}
 \end{aligned}$$

$$\begin{aligned}
 \text{d) Vol.} &= \pi \int_{-1}^1 (x+2) dx \\
 &= \pi \left[\frac{x^2}{2} + 2x \right]_{-1}^1 \\
 &= \pi \left[\left(\frac{1}{2} + 2 \right) - \left(\frac{1}{2} - 2 \right) \right] \\
 &= \pi (2\frac{1}{2} - -1\frac{1}{2}) \\
 &= 4\pi \text{ u}^3
 \end{aligned}$$

$$\begin{aligned}
 (9) \text{ a) } x &= 0.76^\circ (\text{1st, 4th quadrants}) \\
 &= 0.76^\circ, 5.52^\circ
 \end{aligned}$$

$$\text{b) i) } y' = 4 \sin^3 x \cos x$$

$$\text{ii) } \frac{1}{4} \sin^4 x + c$$

$$\begin{aligned}
 \text{c) } \int_3^7 f(x) dx &\doteq \frac{1}{3} (2 + 4 \times 5 + 1) + \frac{1}{3} (1 + 4 \times 3 + 4) \\
 &= \frac{1}{3} (2 + 20 + 2 + 12 + 4) \\
 &= \frac{1}{3} (40) \\
 &= 13\frac{1}{3}.
 \end{aligned}$$

⑨ d) i) graphs intersect when $x^2 = 2x$

$$\therefore x^2 - 2x = 0$$

$$x(x-2) = 0$$

$$\therefore x = 0 \text{ or } 2$$

$$\therefore A \text{ is } (2, 4)$$

$$\text{ii) Area} = \int_0^2 (2x - x^2) dx \quad \text{OR} \quad \left| \int_0^2 (x^2 - 2x) dx \right|$$

$$= \left[x^2 - \frac{x^3}{3} \right]_0^2$$

$$= (4 - 2\frac{2}{3}) - (0 - 0)$$

$$= \frac{1}{3} u^2$$

e) (using x axis) Area = 1x1 square - area to y axis

$$= 1 - \left| \int_0^1 (x^2 - 2x) dx \right|$$

$$= 1 - \left| \left[\frac{x^3}{3} - x^2 \right]_0^1 \right|$$

$$= 1 - \left| \left(\frac{1}{3} - 1 \right) - (0 - 0) \right|$$

$$= 1 - \left| -\frac{2}{3} \right|$$

$$= \frac{1}{3} u^2$$

(some Ext 1 boys may use the y axis)

$$y+1 = x^2 - 2x + 1$$

$$(x-1)^2 = y+1$$

$$x-1 = \pm \sqrt{y+1}$$

$$\therefore x = 1 \pm \sqrt{y+1}$$

(test, choose $-\sqrt{}$)

$$\therefore x = 1 - \sqrt{y+1}$$

$$\therefore \text{area} = \int_{-1}^0 (1 - (y+1)^{\frac{1}{2}}) dy$$

$$= \left[y - \frac{2}{3} (y+1)^{\frac{3}{2}} \right]_{-1}^0$$

$$= (0 - \frac{2}{3}) - (-1 - 0)$$

$$= -\frac{2}{3} + 1$$

$$= \frac{1}{3} u^2 \text{ as above.}$$